

Tutorial I

1. Consider the two kinds of limits (taken limits of x and y at once or one by one) of the following functions at $(0,0)$. If exists, find its value.

$$1.1. \quad f(x, y) = \frac{x^2 y^2}{x^2 y^2 + (x - y)^2};$$

$$1.2. \quad \text{Let } D = \{(x, y) \mid x + y \neq 0\}, \quad f(x, y) = \frac{x - y}{x + y}, \quad (x, y) \in D;$$

$$1.3. \quad f(x, y) = \begin{cases} x \sin \frac{1}{y}, & xy \neq 0 \\ 0, & xy = 0. \end{cases}$$

$$2. \quad \text{Let } f(x, y) = \begin{cases} \frac{\ln(1 + xy)}{x}, & x \neq 0; \\ y, & x = 0. \end{cases} \quad \text{Is it continuous in its domain?}$$

$$3. \quad \text{Let } f(x, y) = \begin{cases} (x^2 + y^2) \sin \frac{1}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases} \quad \text{Prove } f(x, y) \text{ is}$$

differentiable at $(0,0)$, but its partial derivatives are not continuous at $(0,0)$.

4. If both partial derivatives of $f(x, y)$ exist, and continuous at $(0,0)$.

If $f'_x(0,0) = 3$, $f'_y(0,0) = 4$. What is the value of $\lim_{t \rightarrow 0} \frac{f(t, t) - f(0,0)}{t}$.

5. Compute the following formula:

5.1. Let $z = z(x, y)$ determined by $x^2 + y^2 + z^2 = a^2$, find the value of

$$\frac{\partial^2 z}{\partial x \partial y}.$$

5.2. Let $z = z(x, y)$ determined by $(z + y)^x = x^2 y$, find the value of

$$\left. \frac{\partial z}{\partial y} \right|_{(3,3)} .$$

5.3. Let $z = z(x, y)$ determined by $xyz + \sqrt{x^2 + y^2 + z^2} = \sqrt{2}$, and $z(1, 0) = -1$, find the value of $dz|_{(1,0)}$.

6. Prove: Let $D \subset \mathbf{R}^2$ nonempty, and $z = f(x, y) \in C^2(D)$. $f''_{xx} + f''_{yy}$ is invariant under the rotation transformation

$$u = x \cos \theta + y \sin \theta, \quad v = -x \sin \theta + y \cos \theta$$

7. Find the Taylor expansion of second order of $\ln(1 + x + y + z)$ at $(x, y, z) = (0, 0, 0)$.

8. Find the shortest distance from $(0, 0, 0)$ to $z^2 = xy + x - y + 4$.